

Module 1

DC Circuit Analysis and Theorems

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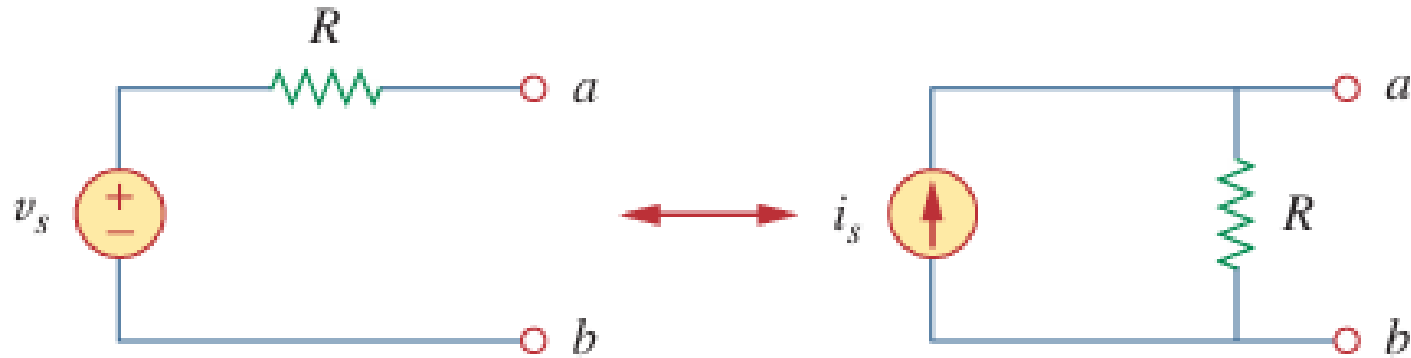
Source Transformation

1. Source transformation is another tool for simplifying circuits.
2. In circuit analysis to be able to substitute a voltage source in series with a resistor for a current source in parallel with a resistor or vice versa.
3. Either substitution is known as a source transformation.

The Source Transformation

A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R , or vice versa.

Source Transformation



1. Note that the arrow of the current source is directed toward the positive terminal of the voltage source.
2. Note that source transformation is not possible when $R = 0$, which is the case with an ideal voltage source. However, for a practical, non-ideal voltage source, R is not equal to 0.
3. Similarly, an ideal current source with $R = \infty$ cannot be replaced by a finite voltage source.

Source Transformation

The two circuits are equivalent—provided they have the same voltage-current relation at terminals a-b.

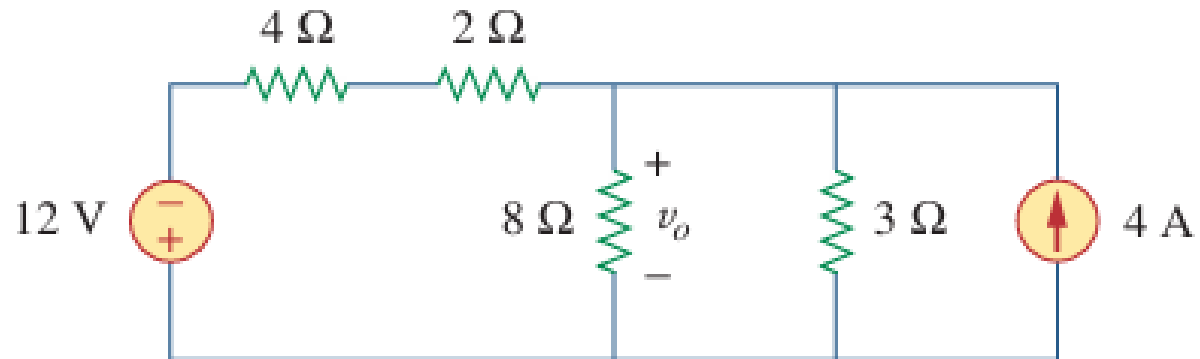
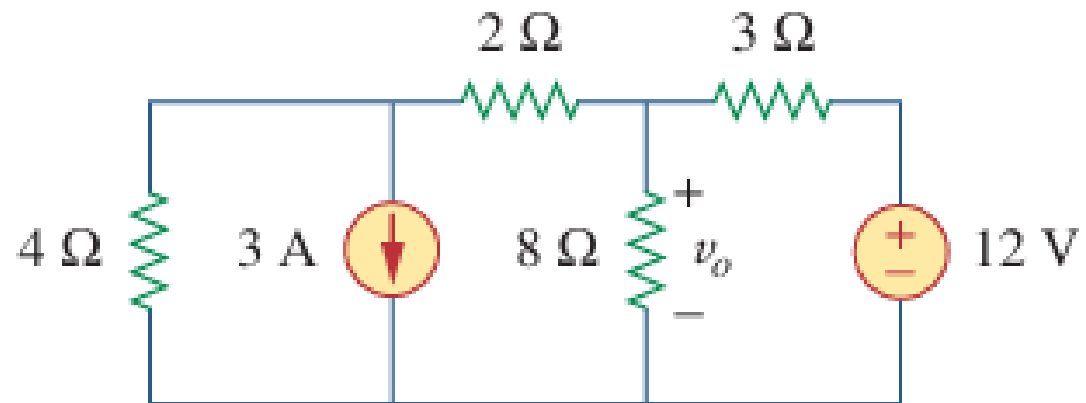
If the sources are turned off, the equivalent resistance at terminals a-b in both circuits is R .

Hence, source transformation requires that,

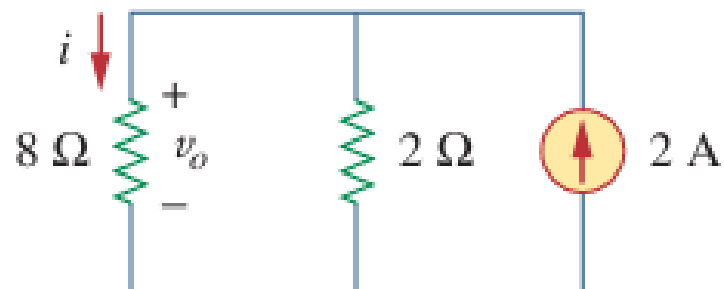
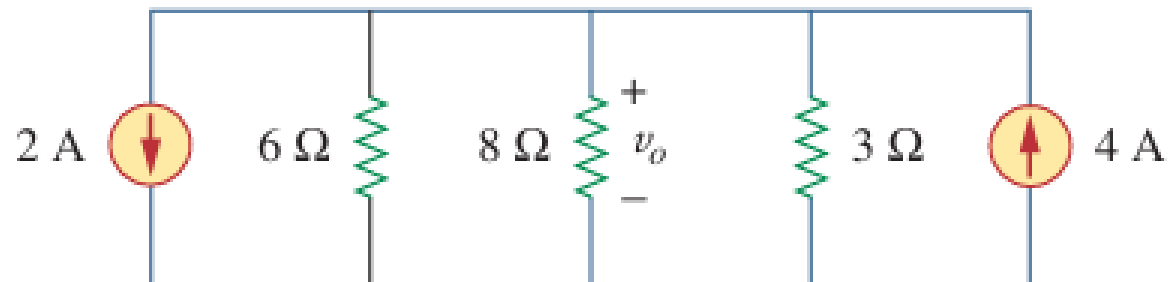
$$v_s = i_s R \quad \text{or} \quad i_s = \frac{v_s}{R}$$

Source Transformation

Use source transformation to find v_o in the circuit.



Source Transformation

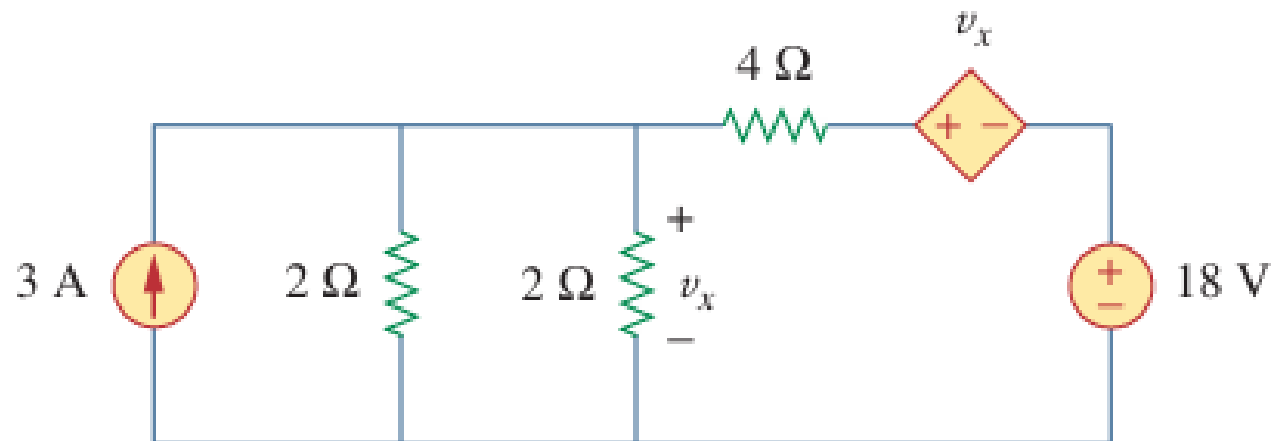
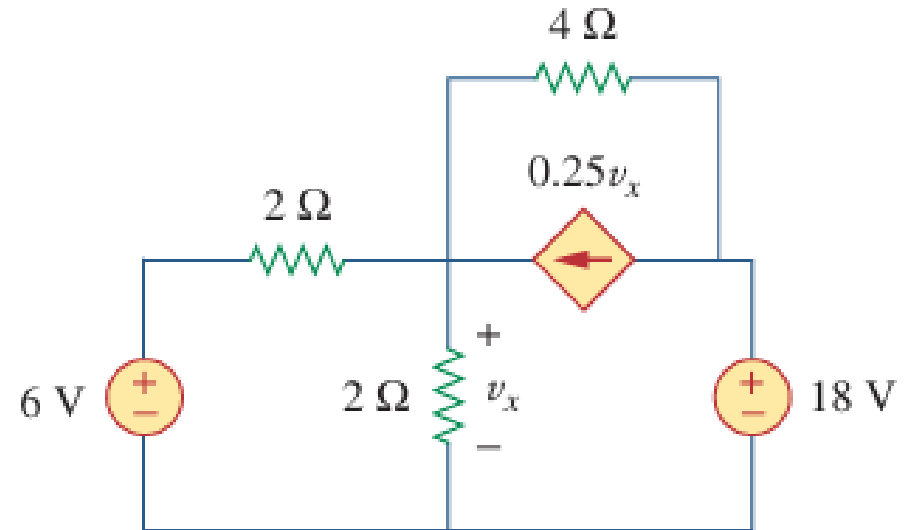


$$i = \frac{2}{2 + 8}(2) = 0.4 \text{ A}$$

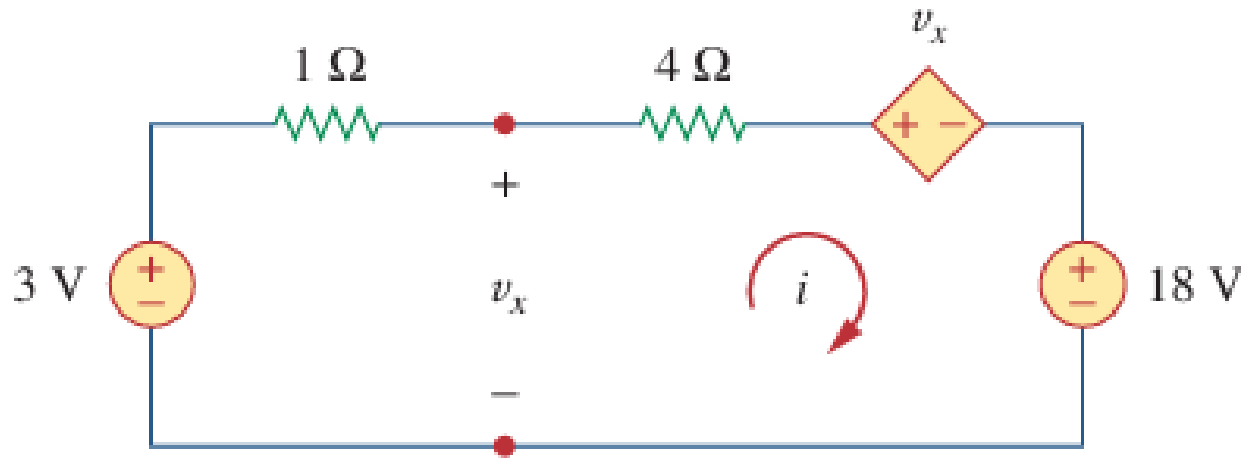
$$v_o = 8i = 8(0.4) = 3.2 \text{ V}$$

Source Transformation

Find v_x using source transformation.



Source Transformation



Applying KVL,

$$-3 + 5i + v_x + 18 = 0 \quad \text{----- 1}$$

Applying KVL to the loop considering v_x ,

$$-3 + 1i + v_x = 0 \quad \Rightarrow \quad v_x = 3 - i$$

Substituting in equation 1,

$$15 + 5i + 3 - i = 0 \quad \Rightarrow \quad i = -4.5 \text{ A}$$

$$v_x = 3 - i = 7.5 \text{ V.}$$

Next class: Thevenin's Theorem